

Healthcare Data Driven Analytics

Hospital Admissions Analysis — Full Project Documentation

Python · MySQL · Power BI Desktop

15,757 Visits · 12,244 Patients · 18.47 Avg Days Stay · 2017–2019

TOOLS	SOURCE	RECORDS	SCOPE
Python · MySQL · Power BI	Cardiology Hospital Admissions Data	15,757 · 59 Cols	12,244 Patients, 2017–2019

01 EXECUTIVE SUMMARY

- **Analyzed 15,757 hospital admission records** from a cardiology-focused hospital dataset spanning 2017–2019 — delivered as two files whose true relationship wasn't what it appeared to be: the field named 'Admission No' turned out to be a reused patient identifier, not a unique visit key, meaning 2,598 patients (21.2%) had more than one recorded visit, one as many as 17.
- **Answered 9 structured clinical questions** covering visit timing, demographics, length of stay, smoking/alcohol disease correlation, and patient outcomes — then reproduced every answer independently in MySQL across a genuinely two-table relational schema (patients + patient_visits) linked by a real foreign key.
- **Built a 3-page, 15-visual interactive Power BI dashboard** with 5 DAX measures and a bucketed analytical view — while navigating a distinctive set of MySQL and Power BI issues specific to this dataset's scale and structure, including a genuine data-corruption bug caught mid-pipeline.
- **Surfaced a counter-intuitive clinical finding worth double-checking against domain literature:** smokers showed a LOWER rate of coronary artery disease than non-smokers (56.1% vs. 70.7%), while diabetes was meaningfully higher in smokers (40.7% vs. 29.4%) — reported as found, not adjusted to match expectation, alongside the honest caveat that a possible selection effect in this specific patient population can't be ruled out from this data alone.

02 PROBLEM STATEMENT

The hospital admissions dataset was distributed as two Excel files — a 56-column clinical record (diagnoses, labs, demographics, dates) and a 2-column name lookup — joined by a field literally named 'Admission No,' which strongly implies a unique key per visit. It isn't: the same admission number recurs for the same patient across separate hospital stays over multiple years, meaning the join is genuinely one-to-many, not one-to-one, and naive deduplication on that field would have silently destroyed thousands of real repeat-visit records. Separately, the date fields governing length of stay contained a real, only-partially-explainable corruption pattern — 7.9% of visits showed a discharge date before the admission date. The goal of this project was to correctly interpret both structural issues, clean the eight mislabeled lab-value placeholders hiding in the clinical columns, and answer 9 concrete questions about admission patterns, risk factors, and outcomes.

03 OBJECTIVES

- **Obj 1:** Correctly diagnose the true relationship between the two source files before assuming Admission No was a unique row identifier.
- **Obj 2:** Investigate the length-of-stay date corruption rigorously enough to make an honest, defensible cleaning decision rather than a convenient one.
- **Obj 3:** Clean 8 lab-value columns using an inconsistent 'EMPTY' text placeholder, plus a separate stray-character placeholder in a 9th column.
- **Obj 4:** Build a genuinely relational two-table MySQL schema reflecting the real one-to-many structure, not a flattened single table.
- **Obj 5:** Reproduce all 9 answers independently in MySQL, catching and fixing a real data-loading bug found through cross-tool validation.
- **Obj 6:** Build a 3-page Power BI dashboard, correctly handling several dataset-scale-specific import and DAX quirks along the way.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Cardiology Hospital Admissions Data (Patient Details + Admission Details)
File Format	2 Excel workbooks, 1 sheet each
Records	15,757 visit rows (56 cols raw → 59 after derived fields) + 12,244 patient rows (2 cols)
True Relationship	Admission No is a reused PATIENT identifier, not a unique visit key — a real one-to-many join
Repeat Visits	2,598 patients (21.2%) have more than one visit; the most frequent patient has 17
Missing Values	8 lab columns + 1 more used inconsistent text placeholders ('EMPTY', '\') instead of true nulls
Date Integrity Issue	1,247 visits (7.9%) show a negative length of stay (discharge before admission) — only partially explained by a day/month-swap theory (721 of 1,247), the remainder left genuinely unresolved
Column Name Defect	'SMOKING ' has a trailing space in its literal column name

Key Columns Used in Analysis

Column	Type	Description
admission_no	Integer	Reused patient identifier — the join key between the two tables
doa / dod	Date	Date of admission / discharge — mixed real-date and text-string cells in the source
los	Integer (derived)	dod – doa; nulled (not dropped) for the 1,247 impossible-negative rows
age / gender / rural	Mixed	Demographics; rural is R/U (rural/urban)
outcome	Text	DISCHARGE (87.3%), EXPIRY (7.0%), DAMA — discharge against medical advice (5.7%)
smoking / alcohol	Binary (0/1)	Lifestyle risk factors
~30 diagnosis flag columns	Binary (0/1)	cad, htn, dm, heart_failure, aki, and 26 more specific conditions
hb, tlc, platelets, glucose, urea, creatinine, bnp, ef	Decimal	Lab values; 'EMPTY' placeholder cleaned to true null

05 TOOLS & TECHNOLOGIES USED

Tool	Platform	Purpose in This Project
Python 3	VS Code	Placeholder cleaning, date-integrity investigation, EDA, visualization
pandas / numpy	Python library	Groupby aggregation, date-swap hypothesis testing, correlation analysis

Tool	Platform	Purpose in This Project
MySQL 8	MySQL Workbench	Two-table relational schema with a real foreign key; SQL validation
Power BI Desktop	Microsoft	3-page dashboard; required specific fixes for FK-surfaced nested columns and TINYINT auto-typing
DAX	Power BI native language	5 measures including a repeat-visit-rate calculation that needed a DAX-limitation workaround

06 DATA CLEANING PROCESS — PYTHON

The most consequential early decision was recognizing what Admission No actually represented, before any cleaning of the clinical columns began.

Step	Action	Result
1	Fix the trailing-space column name ('SMOKING ' → 'SMOKING')	Column now referenceable without a KeyError
2	Replace 'EMPTY' with true null across 8 lab columns; cast to numeric	hb, tlc, platelets, glucose, urea, creatinine, bnp, ef cleaned
3	Replace a stray '\ ' placeholder in CHEST INFECTION with null	9th placeholder-cleaned column, different convention from the other 8
4	Parse D.O.A/D.O.D to true datetime (mixed real-date/text-string cells)	Consistent date type across all 15,757 rows
5	Compute LOS; null out (don't drop) the 1,247 impossible-negative rows	92.1% of visits retain a usable LOS value; full clinical record preserved for the rest
6	Derive Weekday (from D.O.A) and a 6-band Age Group	Two new analysis-ready categorical fields
7	Rename all 59 columns to SQL-safe snake_case	MySQL-ready schema, e.g. 'D.O.A' → doa

07 SQL / MYSQL — VALIDATION LAYER

A genuinely two-table schema reflecting the real relationship: patients (12,244 rows, the parent) and patient_visits (15,757 rows, the child), linked by a real foreign key on admission_no — not a flattened single table, because the data itself is one-to-many.

A Real Bug Caught by Cross-Tool Validation

Running the same LOS query in both Python and MySQL surfaced a genuine discrepancy: Python's average was 18.47 days, MySQL's was 17.00. Root cause: under the relaxed sql_mode needed for the date/text columns to import, MySQL loads an empty CSV field into a numeric column as 0, not NULL — so all 1,247 nulled-out LOS values silently became zeros, dragging the average down. The same risk existed for 9 other nullable numeric columns that hadn't been queried yet. Fix: re-imported with explicit NULLIF(@raw,) on every nullable numeric column, not just the one that happened to get caught.

Analytical VIEW

A vw_patient_visits_summary view buckets visits by age_group, gender, smoking, alcohol, outcome, and rural, pre-computing avg_los and key disease-prevalence rates — added after an initial gap where the view was skipped, since a single flat table plus a view is the pattern used across this project series.

08 DAX MEASURES CREATED

Measure	DAX Formula	Result
Total Visits	COUNTROWS(patient_visits)	15,757
Total Patients	DISTINCTCOUNT(patient_visits[admission_no])	12,244
Avg LOS	AVERAGE(patient_visits[los])	18.47 days
Median LOS	MEDIAN(patient_visits[los])	5 days
Mortality Rate %	DIVIDE(CALCULATE([Total Visits], outcome="EXPIRY"), [Total Visits])*100	7.01

Design note: DAX's native MEDIAN() function made the median calculation trivial — the equivalent SQL query needed a manual window-function workaround, since MySQL has no built-in median. A 6th measure, Repeat Visit Rate %, hit a genuine DAX limitation (SELECTCOLUMNS inside an IN filter isn't permitted as a table filter expression) and needed rewriting with FILTER(SUMMARIZE(...)) instead — documented as its own challenge below.

09 DASHBOARD VISUALS BUILT (15 Visuals, 3 Pages)

Page 1 — Visit Overview

#	Visual Type	Fields Used	Insight Delivered
1-4	KPI Cards	Total Visits / Total Patients / Avg LOS / Median LOS	15,757 visits, 12,244 patients, 18.47/5 day stay split
5	Bar Chart	X: weekday (Mon → Sun) / Y: Total Visits	Monday leads at 2,626; Sunday lowest at 1,684
6	Bar Chart	X: age_group / Y: Total Visits	61–75 dominates at 6,535 (41.5% of all visits)
7	Donut	Legend: gender / Values: Total Visits	63.4% Male / 36.6% Female
8-9	Slicers	gender, rural	Cross-filters all Page 1 visuals

Page 2 — Disease & Risk Factors

#	Visual Type	Fields Used	Insight Delivered
1	Clustered Bar	cad/htn/dm/heart_failure/aki prevalence, Legend: smoking	CAD lower in smokers (56.1% vs 70.7%); DM higher (40.7% vs 29.4%)
2	Clustered Bar (duplicated)	Same 5 diseases, Legend: alcohol	Near-identical pattern to smoking — the two factors track closely here
3-4	Slicers	smoking, alcohol	Filters both disease comparison charts

#	Visual Type	Fields Used	Insight Delivered
1	Donut	Legend: outcome / Values: Total Visits	DISCHARGE 87.3%, EXPIRY 7.0%, DAMA 5.7%
2	Stacked Bar	X: rural / Legend: outcome	Outcome rates are nearly identical between rural and urban patients
3	Table	name, admission_no, visit count (>1 only)	Surfaces the repeat-patient story directly, sorted descending, max 17 visits
4	Slicer	outcome	Filters the donut and stacked bar

10 KEY INSIGHTS & RESULTS

A. Admission Patterns

- **Monday is the busiest admission day** (2,626 visits) and Sunday the quietest (1,684) — a ~56% gap consistent with a typical elective-plus-emergency hospital rhythm.
- **This is an adult, cardiology-skewed population:** the 61–75 age band alone accounts for 41.5% of all visits; patients under 18 are rare (73 visits, 0.5%).

B. Length of Stay

- **Median (5 days), not mean (18.47 days), is the number to lead with:** the distribution is heavily right-skewed by a genuine long tail of complex cases, so the mean overstates a 'typical' stay.

C. Risk Factors

- **A genuinely counter-intuitive result, reported as found:** CAD prevalence is LOWER in smokers than non-smokers (56.1% vs. 70.7%) — flagged for domain review rather than silently adjusted, since it may reflect a real selection effect in this specific referral population rather than a general clinical relationship.
- **Diabetes moves in the expected direction:** 40.7% in smokers vs. 29.4% in non-smokers — and alcohol shows an almost identical pattern to smoking across all 5 tracked conditions, suggesting the two risk factors are themselves correlated in this population.

D. Outcomes & Repeat Visits

- **87.3% of visits end in a standard discharge;** 7.0% end in death (EXPIRY) and 5.7% in discharge against medical advice (DAMA) — rates that hold nearly steady between rural and urban patients.
- **21.2% of patients are repeat visitors,** one as many as 17 times over the 2017–2019 window — a real chronic-care population within the broader admission data, not a data artifact.

11 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — A Join Key That Wasn't What It Claimed to Be

Problem: 'Admission No' strongly implies a unique identifier per hospital stay, but is actually a reused patient ID — a naive `.drop_duplicates()` on it would have destroyed 3,513 genuine repeat-visit rows.

Solution: Investigated a specific repeated value before assuming it was dirty data, found 17 rows with entirely different dates and diagnoses for the same ID — confirming a real one-to-many structure, then built the schema (and the Power BI relationship) to match it correctly.

Challenge 2 — A Date Corruption Pattern That Was Only Partially Explainable

Problem: 1,247 visits showed a negative length of stay. A day/month-swap hypothesis (common with international date format mismatches) looked promising but only resolved 721 of the 1,247 cases when tested directly.

Solution: Resisted the temptation to apply a partial fix that would be right 58% of the time and silently wrong the rest — nulled the value out instead of guessing, and reported the exclusion rate transparently rather than presenting a falsely precise average.

Challenge 3 — Two Different Null Placeholder Conventions in the Same File

Problem: 8 lab columns used the string 'EMPTY'; a 9th column (CHEST INFECTION) used a stray backslash instead — a single blanket replace() would have missed one or the other.

Solution: Checked each column's actual non-numeric values individually during assessment rather than assuming a single placeholder convention applied dataset-wide.

Challenge 4 — Empty Numeric Fields Loading as Zero, Not Null, in MySQL

Problem: Cross-checking the same LOS query in Python (18.47) against MySQL (17.00) surfaced a real, silent bug — empty CSV fields were loading as 0 under relaxed sql_mode, corrupting the average.

Solution: Re-imported with explicit NULLIF(@raw,"") on every nullable numeric column, not only the one caught by the discrepancy — 9 other columns had the identical latent risk before any query had exposed it.

Challenge 5 — A DAX Table-Filter Limitation

Problem: The Repeat Visit Rate % measure used SELECTCOLUMNS inside an IN filter within CALCULATE — DAX rejected it as an invalid table filter expression.

Solution: Rewrote the logic using FILTER(SUMMARIZE(...)) to group and count visits per patient directly, avoiding the IN/SELECTCOLUMNS combination entirely — same result, no parser conflict.

Challenge 6 — Power Query Auto-Surfacing Related Tables as Nested Columns

Problem: Loading the two FK-linked MySQL tables into Power Query auto-added a nested 'Table' column into each, embedding the entire related table inside every row — not usable directly, and duplicating data already loaded as its own clean query.

Solution: Removed the nested column from both queries before Close & Apply; the real relationship gets built afterward in Model view, not through this nested-column mechanism.

12 SKILLS LEARNED & TAKEAWAYS

- Verifying what a join key actually represents by inspecting real repeated rows, rather than trusting a field's name.
- Knowing when a partial algorithmic fix (the day/month-swap theory, correct 58% of the time) is worse than an honest null — precision that's sometimes wrong is more dangerous than an acknowledged gap.

- Auditing placeholder conventions column-by-column instead of assuming one convention applies to an entire file.
- Cross-tool validation as a bug-finding method, not just a confirmation step — the MySQL zero-vs-null bug was only caught because the same number was computed twice, in two different tools, and compared.
- Generalizing a fix beyond the single column where a bug was caught — 9 other columns shared the identical risk and were fixed proactively, not reactively.
- Working around real DAX and Power Query platform limitations (a table-filter parser restriction, FK-driven nested columns) rather than treating every unexpected editor behavior as a personal mistake.

13 CONCLUSION

- **Correctly identified the true one-to-many relationship** hiding behind a misleadingly-named 'Admission No' field before it could silently corrupt 3,513 real visit records through naive deduplication.
- **Investigated a genuine date-corruption pattern honestly**, choosing to null 1,247 unresolved values rather than apply a partial fix that would be confidently wrong 42% of the time.
- **Caught and fixed a real MySQL data-loading bug through cross-tool validation**, then proactively generalized the fix to 9 other at-risk columns before they produced their own silent errors.
- **Delivered a 3-page, 15-visual Power BI dashboard** on a genuinely relational two-table schema, working through platform-specific DAX and Power Query limitations along the way rather than working around the data model.
- **Most notably, reported a counter-intuitive clinical finding as found:** lower CAD prevalence in smokers than non-smokers, flagged clearly for domain review rather than adjusted to fit prior expectation — the data gets the final word, with the appropriate caveat attached.

KPI Summary

Metric	Value	Metric	Value
Total Visits	15,757	Total Patients	12,244
Repeat-Visit Patients	2,598 (21.2%)	Max Visits (1 Patient)	17
Avg LOS	18.47 days	Median LOS	5 days
Gender Split	63.4% M / 36.6% F	Top Age Group	61–75 — 41.5% of visits
Busiest Weekday	Monday — 2,626 visits	Quietest Weekday	Sunday — 1,684 visits
Discharge Rate	87.3%	Mortality Rate	7.01%
CAD (Smokers vs Non)	56.1% vs 70.7%	DM (Smokers vs Non)	40.7% vs 29.4%
Negative-LOS Rows (nulled)	1,247 (7.9%)	Lab Placeholder Columns Fixed	9